OCP GLOBAL SUMMIT 2026 • XANADU

用光子把量子计算机做大

Jonah Force Hill（Xanadu 业务拓展与政府关系总监）

从 Aurora 样机到量子数据中心

时长 20 分 43 秒

来源 youtube.com/watch?v=ibZrZboxSnU

用光子把量子计算机做大

Jonah Force Hill（Xanadu 业务拓展与政府关系总监）OCP Global Summit 2026 演讲 • 图文实录

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这是一份**逐段实录**，不是要点综述。每一节遵循同一结构：

- **图片**：演讲当时投影的幻灯片，按时间顺序
- **中文译文**：讲者在这一页停留期间所说的话，逐句翻译，不作删减
- **英文原句**：同段话的英文原文，供核对与引用
- **小结**：整理者对这一段的一句话归纳，与讲话内容明确区分

原文由 YouTube 自动字幕转写，已去除 um、uh、you know 等口头填充词，并修正明显的转写错误：**Jonah Force Hill**（字幕作 Jonathon）、**Xanadu**（字幕偶作 Xanadu Xanadu）、**error correction**（字幕作 air correction）、**thin-film lithium niobate**（字幕作 TiFlin）、**PNR**（光子数分辨探测器）、**qubit count**（字幕作 cubic count）。

凡口播数字与幻灯片标注不一致，正文采用幻灯片值并显式注明差异；仅存在于口播、幻灯片上无对应的数字，标注为「口头表述」。

本场幻灯片使用**分步构建**（build）：同一页在讲解中逐块显示。文中图片取自该段讲解结束时的画面，能完整反映讲者当时指着的内容。

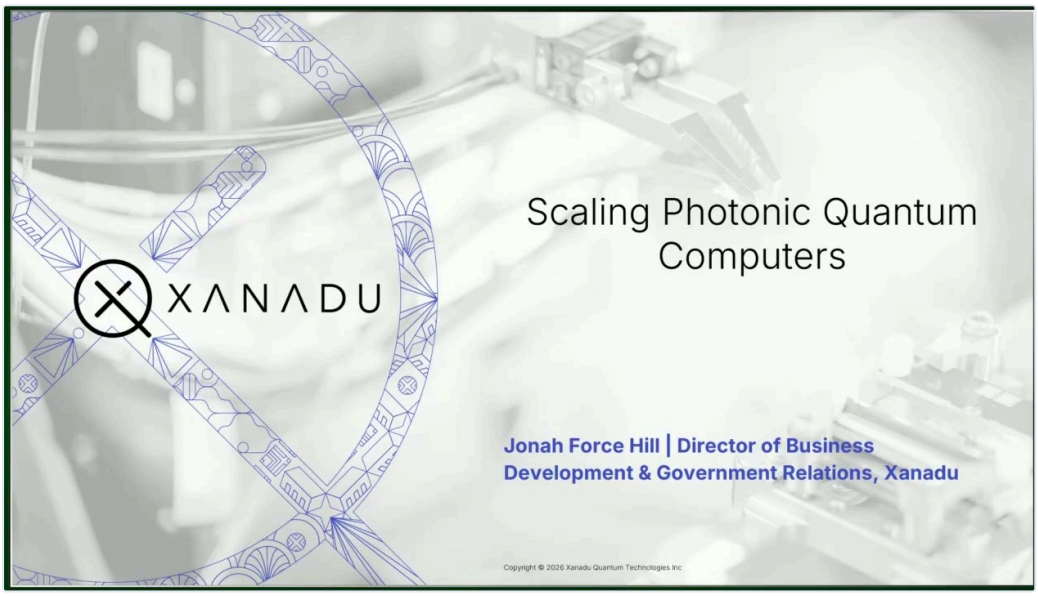
视频信息

项目	内容
标题	Scaling Photonic Quantum Computers
讲者	Jonah Force Hill，Xanadu 业务拓展与政府关系总监（Director of Business Development & Government Relations）
场合	OCP Global Summit 2026 · Executive Session
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项目	内容
频道	Open Compute Project（YouTube）
链接	https://www.youtube.com/watch?v=ibZrZboxSnU

第一部分　为什么是光子

01　开场：十年，一条路线



Scaling Photonic Quantum Computers：Xanadu 封面页，讲者 Jonah Force Hill，Director of Business Development & Government Relations

好，测试，测试。大家好，非常感谢今天请我来。

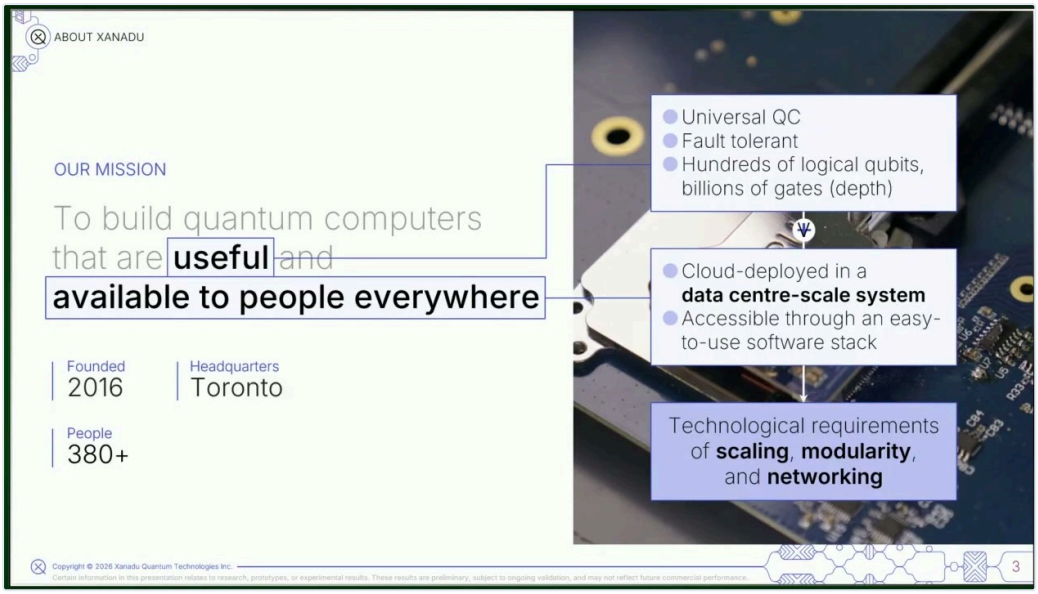
我叫 Jonah Force Hill，是 Xanadu 的业务拓展与政府关系总监。这个角色主要是拓展 Xanadu 的商业合作、政府合作和研究合作。

今天我要讲的是 Xanadu 通往容错量子计算的路径——以光子作为我们的原生技术。

> All right, testing, testing. Hi everybody. Thanks so much for having me here today. My name is Jonah Force Hill. I am Xanadu's director of business development and government relations. That means my role is principally about expanding Xanadu's commercial partnerships, government partnerships, and research partnerships. I'm going to be talking today about Xanadu's path to fault-tolerant quantum computing, leveraging photonics as our native technology.

> **小结**　开场就把立场说清楚了：不是「我们也做量子计算」，而是「我们做的是光子路线的容错机」——后面所有论证都围绕「光子为什么能规模化」展开。

02 「有用」和「人人可用」是什么意思



About Xanadu：使命为「打造对人有用的量子计算机，并让人人都能用上」；2016 年成立、总部多伦多、380+ 人；右侧把使命拆成「通用量子计算 + 容错 + 数百逻辑比特、数十亿门」到「云上数据中心级部署 + 易用软件栈」，落点是规模化、模块化与组网

Xanadu 刚刚过了十周年，所以大概算是量子计算领域的「老卫队」之一了。

我们有一个十年没变的使命：打造对人有用的量子计算机，并让人人都能用上。这句话到底什么意思，我马上讲。

公司总部在加拿大多伦多，过去十年成长很大，尤其是最近三年——全职员工数量几乎翻了一倍。我刚加入的时候，大约 70% 的全职员工是物理、计算机科学或电子工程的博士。随着团队扩张，这个比例下降了一些，但它仍然是这家公司最持久的特征。

说到「有用」，意思是我们需要一台通用的量子计算机，能调用所有可用的门，从而实现尽可能多的算法。

而这意味着容错。在我们看来，在能够真正抑制、并最终纠正这类系统固有的错误之前，不会出现重大的商业成功。

这意味着我们需要数百个逻辑比特和数十亿个门。

而要做到「人人可用」，光有一台别人碰不到的机器是不够的。它必须真的能被所有人使用——所以我们设想，未来的量子计算会部署成云上数据中心规模的系统，并且能通过一套易用的软件栈访问。

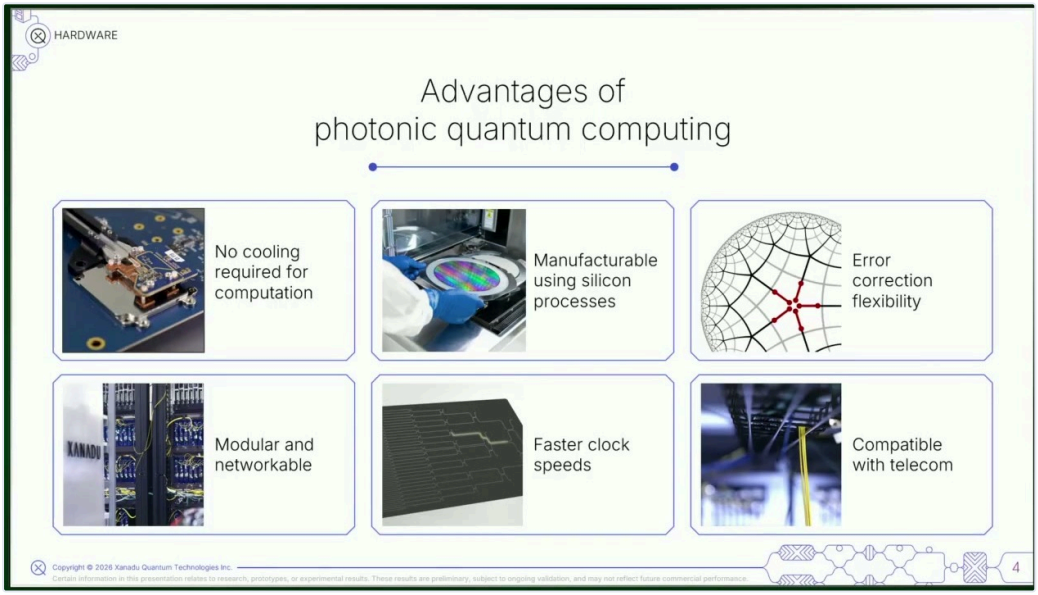
今天我要讲的主要是 Xanadu 的硬件路线，但我们同样花了大量时间和精力建设量子软件生态，主要是我们的 PennyLane 软件包——它是目前最领先的量子计算编程包之一。

> So, Xanadu just celebrated our 10-year anniversary, so we are, I guess, one of the older guards for quantum computing. We have a pretty clear mission that's endured over those 10 years, which is to build quantum computers that are useful and available to people everywhere. I'll get into what that actually means in a second. The company's headquartered in Toronto, Canada, and has grown significantly over the past 10 years, particularly in the last three, where we've nearly doubled the number of full-time staff. When I joined, something like 70% of the full-time employees were PhDs in physics or computer science or electrical engineering. As we've expanded the team, that overall percentage has dropped

> some, but that still remains really the enduring feature of the company. So, when we talk about useful, that means we need a universal quantum computer that's really able to leverage all the available gates to be able to implement the widest variety of algorithms possible. That means fault tolerance. We are not, in our view, going to have major commercial success until we're able to really mitigate and then ultimately correct the errors that are inherent in these systems. That means we're going to need hundreds of logical qubits and billions of gates. And to make it available to people everywhere, it's not sufficient to just have a machine that is inaccessible. It really needs to be available to people everywhere, so we envision quantum computing in the future will be deployed in a cloud data center-scale size system. And that needs to be accessible through an easy-to-use software stack. And I'm going to be talking principally about Xanadu's hardware approach today, but we have spent as much time and energy building up the quantum software ecosystem, principally through our PennyLane software package, which is one of the leading quantum computing programming packages available today.

> **小结**　「有用」被拆成了三个可验证的条件：通用、容错、云上可得。前两条是技术指标，第三条决定了这台机器最终长成数据中心而不是实验室设备。

03　光子路线的六条优势



Advantages of photonic quantum computing：六个方框——计算无需制冷、可用硅工艺制造、纠错方案灵活、模块化可组网、时钟速度更快、与电信兼容

光子量子计算有一批固有的优势。

计算不需要制冷。 探测环节确实需要低温，但从制备特定量子态到运行门，所有计算都在室温下完成——这让我们能大幅压缩整机的占地，尤其是在制冷和供电需求上。

硅光子可以用标准或改造过的硅工艺制造，这意味着我们可以用上全世界最先进的那批晶圆厂。

因为这套系统用的是**簇态（cluster state）**——存在于时间和空间中的三维网格态——我们的纠错灵活性非常高。我们并不受某种纠错方案的限制，可以把业界正在开发的最先进纠错方案，原样、原生地实现进我们的架构。

模块化、可组网。 光子天然跑在光纤上，这意味着一旦你有一个包含若干比特的模块，就可以把它往外扩，几乎没有上限。

门速度。 我们的门速度很快，运行时间因此更短。

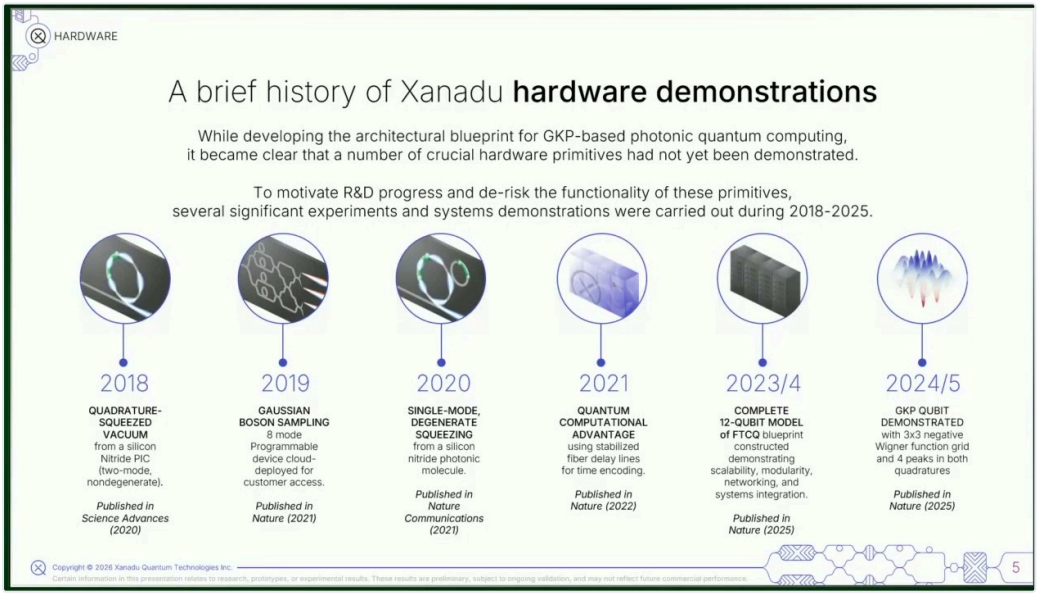
与电信兼容。 就像我说的，我们可以直接在光纤上跑，不需要对信号或波长做任何转换或改造。

> So, there's a lot of advantages that are inherent to photonic quantum computing. No cooling is required for the computation. Now, there is cryogenic that's required for some of the detection, but all the computation — from the generation of particular states to running gates — all of that is done at room temperature, which allows us to really reduce the overall footprint of the machine, especially on the cooling and power requirements. Silicon photonics is manufacturable using standard and modified silicon processes, but it allows us to leverage some of the most advanced fabs around the world. Because this is a system that uses cluster states, which are 3D states, 3D grid states, that exist in time and space, we have tremendous error correction flexibility. So, we're not really limited by any error correction restrictions. We can take the most advanced error correction schemes that are being developed and easily implement them into our architecture natively. Modular, networkable: photonics, because it runs natively over fiber optics, means that once you actually have a module that has a certain number of qubits within it, you can easily scale that out really indefinitely. Photonics on gate speed: we have a very fast gate speed, which allows for faster run times. And then, compatible with telecom. As I mentioned, we have the ability to run over fiber optic cables without having to do any kind of transduction or modification of the signals that we're using or the wavelengths.

小结 六条优势可以归成一句：光子路线把「规模」变成了工程问题而不是物理问题——不用制冷、能上现有晶圆厂、能靠光纤往外堆。

第二部分 用演示项目逐个证明子系统

04 2018–2025：六个演示，六篇 Nature



A brief history of Xanadu hardware demonstrations：2018 压缩真空（Science Advances 2020）、2019 高斯玻色采样（Nature 2021）、2020 单模简并压缩（Nature Communications 2021）、2021 量子计算优势（Nature 2022）、2023/4 十二比特 FTQC 蓝图模型（Nature 2025）、2024/5 GKP 比特演示（Nature 2025）

Xanadu 一直采取的做法是：**容错始终是目标。**

而要做到容错，我们需要一系列演示项目，证明我们正在搭建的每一个子系统和部件都按预期工作，证明我们设计的架构确实能行、并且会行。

所以从 2018 年起，我们几乎每年发布一个新的演示项目，并配一篇论文，其中大部分发表在 Nature 上——这也说明这项工作受到的审视有多严格。

从光源和高斯玻色采样开始——我们要确保这些系统能在云上使用，让用户可以上手、可以测试。而且我们需要那份额外的验证，来确认「这东西是能成的」。

稍等。

2022 年，我们做了量子计算优势的演示。那次我们解了一个很冷门的数学问题，没有什么商业价值，但如果用当时最快的超级计算机，大概要 700 万年。

那是全球第三个同类演示——前面是谷歌，然后是中国的那个中心，做过类似的量子优势实验。

同样，它没有特别实际的用途，但它证明了：人们期待这类机器带来的加速是真实的、正在到来的。

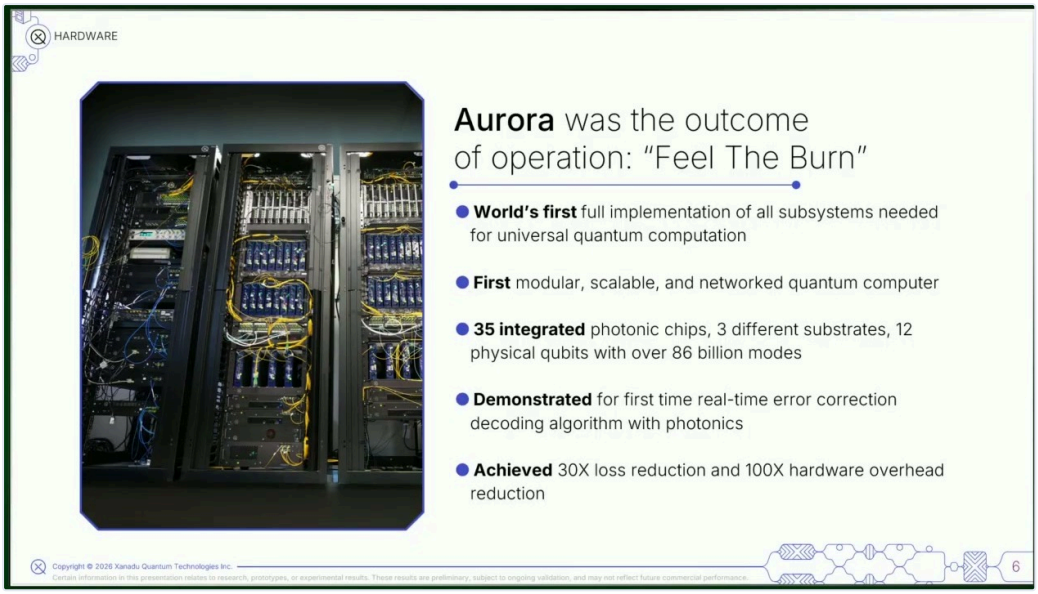
最后是**片上 GKP 比特**。那是第一次在芯片上生成 GKP 比特。

而每一块拼图，都在说明：这些正在被造出来的子系统，一旦拼到一起，就会按预期工作。

> So, Xanadu has really taken the approach that fault tolerance has always been the goal. And to achieve fault tolerance, we needed to have a series of demonstrators to show that all of the subsystems and component parts that we're building work as intended, and that the architecture that we're designing, in fact, can work and will work. And so, since 2018, we've released about every year a new demonstrator project that's been paired with a publication, most of which have been in Nature, which really shows the seriousness with which this work is being reviewed. Starting with the sources and Gaussian boson sampling — we wanted to make sure that these systems were available on the cloud so that users could play with them, test them. And again, we needed that additional validation to say, this is something that's going to work. One second. So, in 2022, we had our quantum computational advantage demonstration. This is where we solved an esoteric math problem that doesn't really have any commercial utility, but was something that would have taken something like 7 million years on the at-the-time fastest supercomputer. That really was the third such demonstration globally, after Google and then the group in — excuse me, the center in China — that had a similar kind of experiment demonstrating quantum advantage. Again, not for any particularly useful use, but again, was showing that the kinds of speedups that one can anticipate from these machines is real and coming. And then finally, we had our GKP qubit on a chip. This was the first time that a GKP qubit had been generated on chip. And really, just each of these pieces showed that those subsystems that are being built, when brought together, will work as intended.

> **小结**　这条时间线的逻辑不是「越做越大」，而是「每年拆掉一个未知数」：光源、采样、压缩、优势、蓝图、比特，每一步都在为最后那台容错机背书。

05　Aurora：一次「烧掉风险」的行动



Aurora was the outcome of operation: "Feel The Burn"：世界首个包含通用量子计算全部子系统的实现；首个模块化、可扩展、可组网的量子计算机；35 颗光子芯片、3 种基板、12 个物理比特、超过 860 亿个模式；首次用光子实现实时纠错解码；损耗降低 30×、硬件开销降低 100×

我想重点讲一下我们的 Aurora 演示。

它来自 2025 年，是一个为期两年的项目，我们把它叫作「Feel the Burn」（烧掉风险）。

它的核心就是把架构里的各种风险逐个烧掉，证明——像之前的那些子系统一样——这些东西能被整合到一起，做成一种可以在数据中心环境里复制的形态。

这也是我们把通用、模块化、可组网量子计算所需的每一个子系统第一次拼在一起的机会。

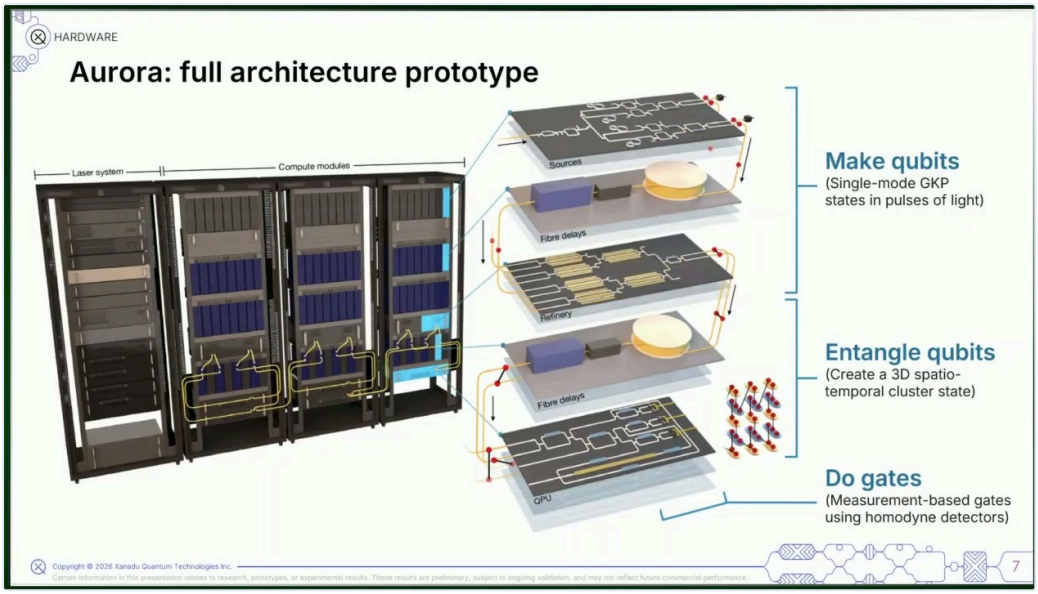
于是我们有了 35 颗光子芯片，产生了 12 个物理比特，并且能跑实时的纠错与解码。

在那段时间里，我们把光损耗和硬件开销都降到了原来的百分之一（幻灯片标注：损耗降低 30×、硬件开销降低 100×）。所以这确实相当可观。

> So, wanted to just highlight a little bit of our Aurora demonstration. This was from 2025. It was a 2-year long project. What we dubbed "Feel the Burn." And this was really about burning down the various risks of our architecture, to show, again, like the subsystems before, that all of these things could be brought together and integrated in a form factor that could be replicable in a data center environment. And this was really an opportunity for us to put together every subsystem that's needed for universal modular networked quantum computation. So, we had 35 photonic chips that generated 12 physical qubits, and we were able to run real-time error correction and decoding. We, over the course of that time, brought down optical loss and our hardware overhead by 100x. So, it was really quite impressive.

> **小结**　Aurora 的价值不在 12 个比特，而在于它第一次证明「所有子系统能同时在场」——这一步过了，剩下的就只是把损耗数字压下去。

06　架构分三步：造比特、纠缠、做门



Aurora: full architecture prototype：左侧激光系统与计算模块机柜，右侧分层爆炸图为 Sources → Fibre delays → Refinery → Fibre delays → QPU，并标注三步——造比特（脉冲光中的单模 GKP 态）、纠缠比特（构建三维时空簇态）、做门（用零差探测器做基于测量的门）

Aurora 是第一个把所需子系统全部包含进来的量子计算系统。

我来讲一下这套架构的结构。左边——抱歉，图有点小——是激光系统；右边是计算本身所需要的全部东西。

光沿着这个堆栈往下走，其实只有三步。

第一步，造比特。 用光源芯片——我们的氮化硅平台——生成量子态。

第二步，纠缠比特。 我们用薄膜铌酸锂开关和光纤延迟线，把这些态连成一个大的纠缠结构。就像我之前说的，我们把它叫作簇态，它跨越空间和时间连在一起。

最后一步，做门。 QPU 芯片对每一个模式做测量，而「选择测什么」，本身就构成了计算。

需要说清楚的是：Aurora 是一个缩比模型。光子损耗——它是核心噪声源，相当于其他量子计算架构里的噪声——仍然远高于容错阈值。

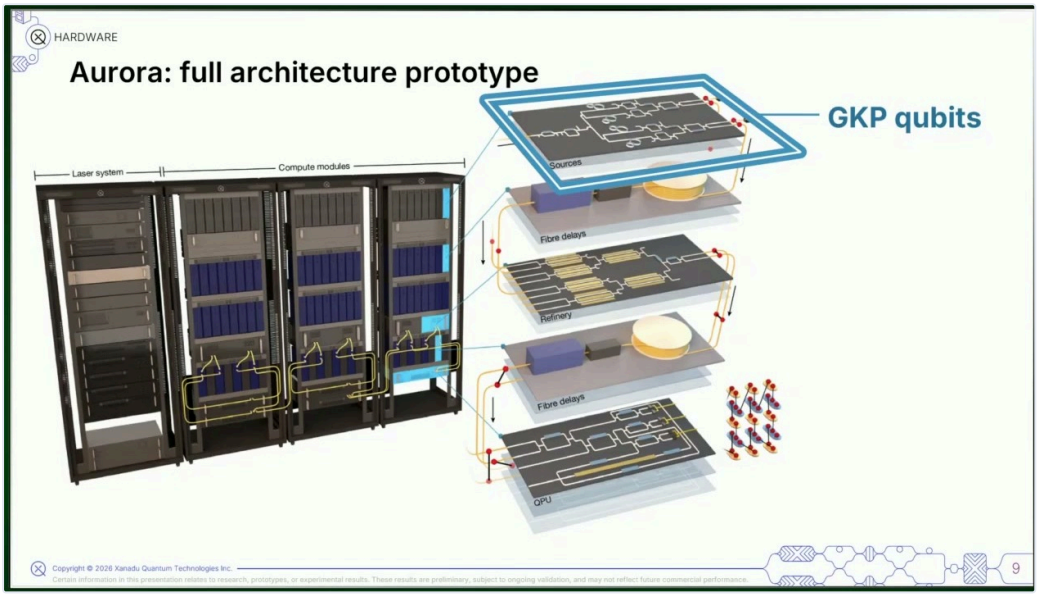
但它证明的是：这些模块能被造出来，能用光纤连起来，能同步，并且能在类似数据中心的环境里连续运行。

> Aurora was really the first quantum computing system that includes every subsystem needed for that. And I'll talk a little bit about the structure of the architecture here. On the left, we have — sorry, it's small here — we've got the laser system, and then on the right, we've got everything that's required for the computation itself. So, the light moves down the stack in really three steps. First, you make the qubits. So, you take source chips, which are our silicon nitride platform, and you create quantum states. Then, second, you entangle the qubits. We use thin-film lithium niobate switches and fiber delay lines to connect those states into a large entangled structure — as I mentioned earlier, we call it a cluster state, which is linked across space and time. And then lastly, you do the gates. So, the QPU chips measure every mode, and the choice of what to measure is really what the computation is comprised of. So, just to be clear, Aurora was a scale model. The losses — photonic losses, which is really the core noise source, it's the analogous to noise in other quantum computing architectures — those are still far above the threshold for fault tolerance. But what it

> proved was that the modules could be built, they could be connected by fiber, they could be synchronized, and they can run continuously in a data center-like environment.

小结　光子路线的计算范式在这一页才露出来：门不是「施加」的，而是「测量」出来的——所以整个架构的重心从操控转移到了测量与反馈。

07　逐层走一遍：激光、光源、复用、测量



Aurora: full architecture prototype（构建后）：同一张架构图，光源层被高亮标注为「GKP qubits」，用于逐层讲解

那我们逐层走一遍。

最左边是**激光系统**。它其实是整台机器的时钟和电源。那束参考光把相位基准带到每一个模块。

光子量子计算本质上依赖干涉，所以每一个到达芯片的脉冲，都必须相对其他所有脉冲有正确的光相位——参考光束就是提供这个的。用数据中心的话说，这就是这台机器的功率与授时分配，只不过是以光的形式送达。

往右走是**光源层**。Aurora 有 24 颗氮化硅芯片。第一步是对光子比特的产生做**预报（heralding）**。

我们用 GKP 比特，但在 Aurora 里没有用——它其实是 GKP 比特的前身，因为当时损耗太高了。

这也是整个系统里唯一需要低温的部分。我们已经从过渡边缘传感器（TES）演进到了 PNR（光子数分辨探测器）。它工作在几毫开尔文，所以相比其他一些系统，挑战还算小。

这个预报是要花时间的，所以光会在光纤延迟线里等一小会儿——一毫秒多一点——直到解码器的结果回来。图上能看到，就是那条光纤延迟线。要是有根教鞭就好了。

好，接下来是**复用**。预报型光源是概率性的：在某个时钟周期里，任何一个光源都可能产出、也可能没产出好的态。复用就是让「成功」以很高的概率发生——我们并行跑多个光源，把最好的那一路切换进计算机。

光随后穿过稳定化的光纤延迟线，同时探测器的结果在被处理。然后它进入我们称为**精炼层（refinery）**的部分，那里是六颗薄膜铌酸锂芯片，配非常快的电光开关。

这条光纤延迟线，正是让整套系统能在数据中心规模上工作的关键。每一盘都是 200 多米长的线圈，却能整齐地装进一个机架模块。机架里的温度变化会让光程长度漂移，声学噪声会带来抖动，但 SFDL（稳定光纤延迟线）会用相位和偏振补偿把这些漂移修掉。

好，最后一部分是上面两层：把复用后的输出连成资源态，供计算消耗。把这些光脉冲在空间和时间上编织在一起，你就得到一个巨大的三维簇态。

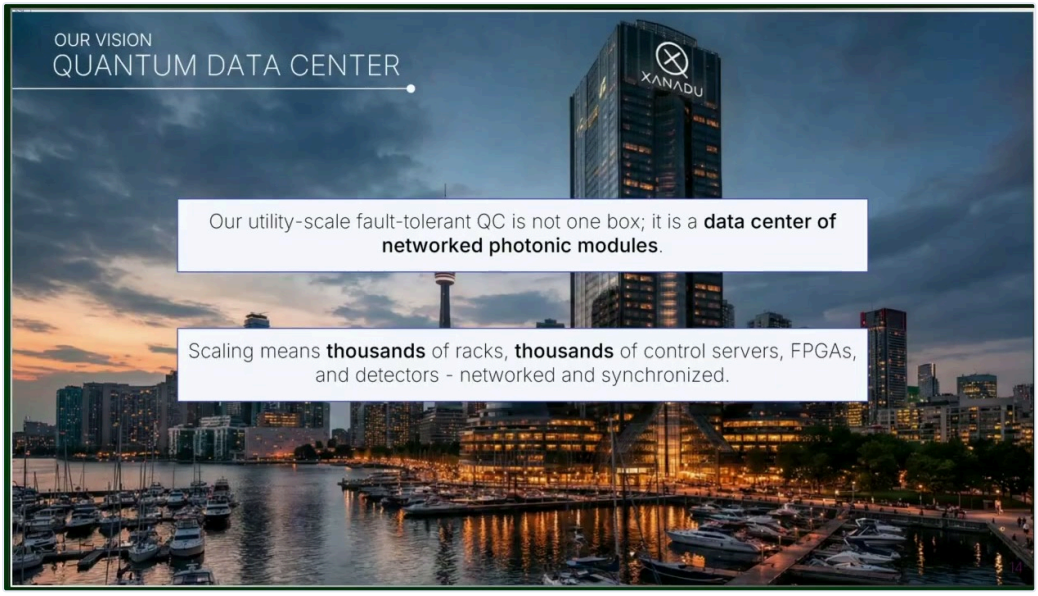
我们有芯片——ASIC 和 SNSPD——实时测量这些光，并用超快反馈来纠错。

> So, just walking through each part. On the far left, you have the laser system. It's really the clock and the power supply of the entire machine. That reference light carries the phase reference to every module. Photonic quantum computing really depends on interference, so every pulse arriving at the chip must have the right optical phase relative to every other pulse. And the reference beams provide that. In data center terms, this is sort of the power and timing distribution for this machine, but delivered optically. Going to the right, this is the sources layer. So, Aurora has 24 silicon nitride chips. The first step is heralding the creation of those photonic qubits. We use GKP qubits, but here in Aurora we did not — it was really a precursor to the GKP qubit, because the losses were too high at the time. This is really the only part of the system that required cryogenics. We have evolved from transition edge sensors to PNRs. And again, this is at a few millikelvin, so it does present fewer challenges than maybe some of the other systems. Now, this heralding, it takes time, so the light waits in the fiber delay line for a little over a millisecond till the decoder results arrive. So that's — you can see in the diagram, that's the fiber delay line there. I wish I had a pointer. All right, so going to the multiplexing. Herald sources are probabilistic, so on a given clock cycle, any one source may or may not have produced a good state. So multiplexing makes this successful with a high degree of probability. We can run sources in parallel and switch the best output into the computer. The light then passes through the stabilized fiber delay lines while the detector results are processed. Then it goes to what we call the refinery, which is six thin-film lithium niobate chips with very fast electro-optical switches. And this fiber delay line is really what makes this work at data center scale. Each of these is an over 200 m long coil, and it fits neatly within a rack module. The temperature changes in the rack make the optical path length drift, and acoustic noise makes it have some jitter, but the SFDLs, the fiber delay lines, correct for that using phase and polarization — correct for the phase and polarization drifts. All right, so the last part is the last two layers connect the outputs to the multiplexing into the resource state that is consumed to perform the computation. By weaving these pulses of light together in space and time, you create a massive three-dimensional cluster state. And we have chips — both ASICs and SNSPDs — that measure this light in real time and use ultra-fast feedback to correct errors.

> **小结**　这一层最像数据中心的不是机柜，而是那条 200 米的光纤线圈：它用「延迟」换「决策时间」，把概率性的光源变成确定性的比特流。

第三部分　从样机到数据中心

08　量子数据中心不是一台机器



Quantum Data Center（渲染图）：多伦多夜景中的 Xanadu 大楼，图上文字为「公用事业级容错量子计算不是一个盒子，而是由光子模块组网构成的数据中心」「规模化的含义是数千个机架、数千台控制服务器、FPGA 与探测器，全部组网并同步」

那么这里的愿景是什么？

这只是一张渲染图。我想确保大家知道这只是渲染图，是我们希望在未来看到的样子。

但我们确实想要建一个大型的量子数据中心，用足够多的这类机架，凑到一台「公用事业级」的机器。

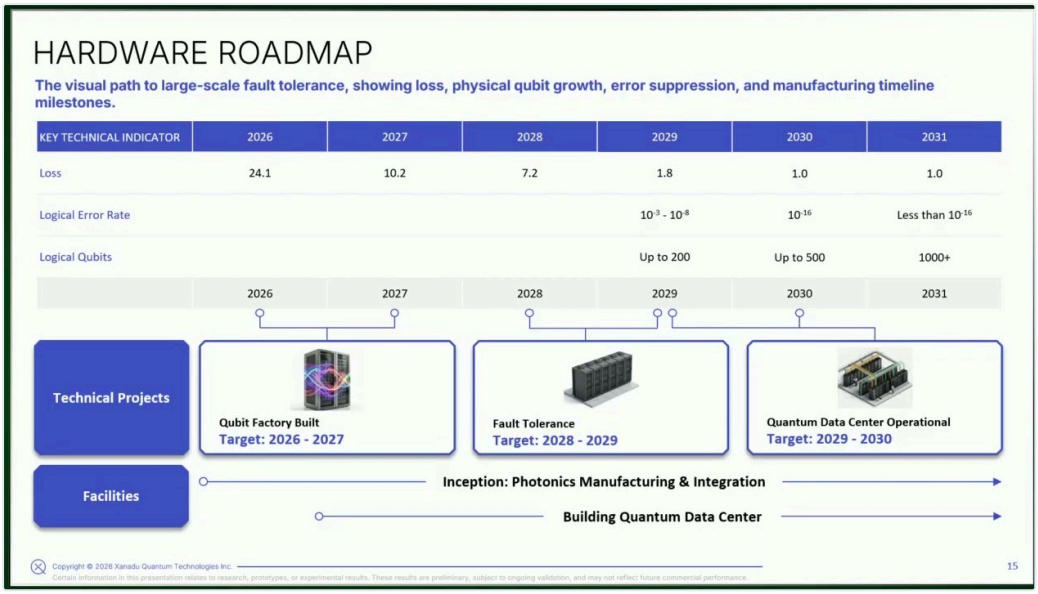
它不是一个你可能在经典计算意义上想象的那种单台计算机。它更像一个数据中心，里面是组网起来的光子模块阵列。

你会有数千个机架、数千台控制服务器、数千块 FPGA 和探测器，全部组网并同步。

> So, what's the vision here? This is just a rendering. I wanted to make sure that folks are aware that this was just a rendering, and just really a vision of what we hope to see in the future. But we would like to have a large-scale quantum data center built with enough of these racks to get to a utility-size machine. And it's not a single computer in the way that maybe you might think of in a classical sense. It is more like a data center with a networked array of these photonic modules. You're going to need thousands of racks, thousands of control servers, thousands of FPGAs and detectors, all networked and synchronized.

> **小结**　这句话是本场与 OCP 接上的地方：光子量子计算机的最终形态是「一个机房」，而不是「一台仪器」——机架、供电、同步、散热这些问题，全都是数据中心的老问题。

09　路线图：把损耗压到阈值以下



Hardware Roadmap：2026–2031 年损耗从 24.1 降到 1.0，2029 年逻辑错误率 10^{-3}–10^{-8}、逻辑比特最多 200，2030 年 10^{-16}、最多 500，2031 年优于 10^{-16}、1000+；技术项目为 Qubit Factory（2026–2027）、Fault Tolerance（2028–2029）、Quantum Data Center Operational（2029–2030）

我们的路线图是几个月前发布的。

你可以看到，核心就是**把所有部件里的损耗降下来**。我们有大约 17 个关键子系统，每一个都需要大幅降损。

这是个艰巨的挑战，但看我们迄今拿到的数字，我们很有信心能达到那些目标。

时间还够吗？五分钟。

所以未来几年，目标就是继续降损，同时改进纠错、改进探测器、改进光纤延迟线，直到所有部件在某个点上汇合——损耗低于阈值，纠错真正开始消除那些困扰系统的错误。

所以，不像很多基于物质体系的量子计算机那样比特数一路往上走——对我们来说，是损耗一路往下走；等损耗足够低，你就可以扩到设想的数千个机架。

回到刚才。我们之所以想参加这个活动、加入这个社区，一个原因是：支撑这类光子架构，需要数据中心的基础设施。

这里有振动和散热问题、有复杂光纤延迟线的集成问题、有时序与同步问题。所以，虽然相比其他路线，我们在规模上更简单，但仍有一批问题要解决——我们指望这个社区的专业能力来帮我们达成。

我就讲到这里，谢谢大家的时间。很乐意回答问题。

> So, we really released our roadmap a few months ago. You can see really it's all about reducing loss in all the components. We have about 17 key subsystems, each of which needs to bring down loss substantially. So, it is a monumental challenge, but given the numbers that we've achieved to date, we feel very confident that we can achieve those numbers. How are we doing on time? Five minutes. And so, over the next several years, the goal is going to be to continue to bring down loss while also improving our error correction, improving our detectors, improving our fiber delay lines, and getting to a point where all of the pieces come together at a point where the loss is below the threshold, and that error correction actually starts to eliminate the errors that plague the systems. So, unlike what you see in a lot

> of the matter-based quantum computers where the qubit count goes like this and it just goes up and up, for us, it's loss goes down and down and down, and then eventually, once loss is low enough, you can scale up to as envisioned thousands of racks. Let's go back. One reason we really wanted to be a part of this event and this community is that there is a need for data center infrastructure to support this kind of photonic architecture. There are vibration and cooling questions, integrating our complex fiber delay lines, there are timing and synchronization issues. So, while compared to maybe other approaches, ours is simpler in terms of scale, there are still a number of problems that we need to solve, and we're going to be looking to this community with the expertise to help us achieve those goals. So, I'll stop there, and thank you all for your time. Happy to take any questions.

> **小结**　路线图的形状很说明问题：别人画的是「比特数往上」，Xanadu 画的是「损耗往下」——因为在这条路线上，规模是损耗的因变量，不是自变量。

第四部分　问答

10　差异化、物理比特开销与真正的瓶颈



Thank you：Xanadu 致谢页，联系方式 jonah.forcehill@xanadu.ai、@XanaduAI

提问：如果从「量子计算即服务」的角度看，市面上已经有好几家了。Xanadu 和其他量子计算公司的差异化在哪里？如果我是一个客户，为什么选 Xanadu，而不是别家的服务？

Jonah Force Hill：就像今天所有量子公司都强调过的，我们还处在早期。所以一旦实现容错，我认为局面会剧烈变化。

但归根到底，是看你的具体需求：你要的是精度、速度，还是价格？这些都是考量。

所以在现阶段，对客户来说它实际上就是一个逻辑比特——问题在于，你带来的问题更适合哪种架构。我想这最终会落到价格，以及某种架构相对其他架构的优势上。

我们的看法是——我想量子行业多数人也同意——需求足够大，会容得下不同的公司提供不同种类的服务和算法，取决于各家公司的专注点。

提问：在这套架构里，多少个物理比特构成一个逻辑比特？有大致概念吗？

Jonah Force Hill：这是个移动靶。随着量子纠错变好、损耗下降，那些开销数字会变。所以在这个时点给你一个硬数字是不负责任的——就像今天讲过的所有架构一样，这个数字会随时间调整。

提问：你提到需要数百个逻辑比特和数十亿个门。那么实现这种规模，最大的瓶颈是什么？

Jonah Force Hill：对我们来说，目前就是损耗。系统里丢掉了太多光子。

你需要让量子纠错既高效、又划算。你不会想要一套物理比特与逻辑比特达到 5000:1 的系统——那在成本上根本站不住。

所以关键是把开销压下来，而要压开销，就得先把损耗压下来。

提问：谢谢，讲得很好。

Jonah Force Hill：谢谢。

> **Q:** So, as you're looking at, I guess, quantum computing as a service, there's a few different ones out there. What's the differentiation between Xanadu and, say, some of the other quantum computing companies, as far as if I'm a customer — why would I go to Xanadu rather than go somewhere else as far as the service is concerned?
> **A:** Yeah, so again, we are, as I think all the quantum companies have emphasized today, still in the early days. So, once we achieve fault tolerance, I think things will change dramatically. But it's really about what is your particular need? Do you need accuracy, speed, price? Those are all going to be considerations. So, I think at this stage it's a logical qubit for all intents and purposes — it's a matter of are the problems that you are bringing to be solved a better fit for one architecture versus another. And I think that really is going to come down to, like I said, price and the kinds of benefits that a particular architecture has over others. So, we're of the mind, and I think most of the quantum industry agrees, that there's sufficient demand that there will be different companies providing different kinds of services and different kinds of algorithms depending on what their focus is at the particular company.
> **Q:** How many physical qubits are making a logical qubit in this architecture? Any idea?
> **A:** Yeah, so that's a moving target. So, as QEC gets better, loss goes down, those overhead numbers are going to change. So, I would be remiss to give you a hard number at this point, because it's like with all the architectures that have come up here today — that adjusts over time.
> **Q:** You mentioned about the hundreds of logical qubits and the billions of gates. So, what's the biggest bottleneck to achieve this kind of scaling?
> **A:** For us right now, like I said, it's loss. We have too many photons lost in the system, and you need to make quantum error correction efficient and you need to make it cost-effective. And you don't want a system that has a 5,000 to 1 physical to logical ratio. It just becomes too expensive a proposition. So, it's really about getting those overheads down, and to do that you need to get loss down. Thank you. That was a good talk. Thank you.

小结　三个问题的答案指向同一件事：这条路线现在只有一个 KPI——损耗。物理/逻辑比是损耗的函数，成本是比值的函数，所以「降价」在光子路线上等价于「降损」。

附录

A. 关键数据核对表

数据项	幻灯片值	口播值	本文采用
成立年份	2016	刚过十周年	一致
总部	Toronto	加拿大多伦多	一致
员工规模	380+	近三年翻倍	一致（口播未给数）
博士占比（讲者加入时）	—	约 70%	口播值
逻辑比特目标	Hundreds of logical qubits	数百个	一致
门数目标	Billions of gates (depth)	数十亿	一致
量子计算优势耗时	—	约 700 万年（当时最快超算）	口播值
Aurora 光子芯片	35 integrated photonic chips	35 颗	一致
Aurora 基板种类	3 different substrates	未给	幻灯片值
Aurora 物理比特	12	12	一致
Aurora 模式数	Over 86 billion modes	未给	幻灯片值
损耗降低	**30×**	**"100x"**	**幻灯片 30×（口播与幻灯片冲突，见注）**
硬件开销降低	100×	100×	一致
光源芯片数	—	24 颗氮化硅	口播值
精炼层芯片数	—	6 颗薄膜铌酸锂	口播值
延迟线长度	—	每盘 200 m 以上	口播值
预报等待时间	—	略多于 1 毫秒	口播值

数据项	幻灯片值	口播值	本文采用
探测器温度	—	几毫开尔文	口播值
关键子系统数	—	17 个	口播值
路线图损耗	24.1 / 10.2 / 7.2 / 1.8 / 1.0 / 1.0（2026–2031）	未给	幻灯片值
逻辑错误率	10^{-3}–10^{-8}（2029）、10^{-16}（2030）、$<10^{-16}$（2031）	未给	幻灯片值
逻辑比特（路线图）	最多 200（2029）、最多 500（2030）、1000+（2031）	未给	幻灯片值
物理/逻辑比	—	「不想要 5000:1」	口播值

> 注：讲者在 08:00 处说「把光损耗和硬件开销都降到了原来的百分之一（100x）」，但同页幻灯片标注为「**30X loss reduction** and 100X hardware overhead reduction」。本文正文按幻灯片标注为 30×，并在译文中保留了讲者的原话与差异说明。

B. 时间轴

时间	内容
00:04	开场：十年，一条路线
00:43	「有用」和「人人可用」是什么意思
03:14	光子路线的六条优势
05:19	2018–2025：六个演示，六篇 Nature
07:41	Aurora：一次「烧掉风险」的行动
08:48	架构分三步：造比特、纠缠、做门
10:35	逐层走一遍：激光、光源、复用、测量
14:13	量子数据中心不是一台机器
15:11	路线图：把损耗压到阈值以下
17:14	问答：差异化、物理比特开销与真正的瓶颈

C. 术语对照

缩写	全称	说明
GKP qubit	Gottesman–Kitaev–Preskill qubit	用振荡器相位空间中的网格态编码的量子比特，对位移误差有天然纠错能力
Cluster state	簇态	大规模纠缠态，本文中指跨越时间与空间的三维网格态，是测量型计算的资源
Squeezed vacuum	压缩真空态	光子路线的基础量子态，噪声在一个正交分量上被压到标准量子极限以下
Gaussian boson sampling	高斯玻色采样	用压缩态与线性光学网络实现的采样任务，用于演示量子优势
PIC	Photonic Integrated Circuit	光子集成电路
TFLN	Thin-Film Lithium Niobate	薄膜铌酸锂，用于高速电光开关
SFDL	Stabilized Fiber Delay Line	稳定光纤延迟线，用延迟换取解码决策时间，并补偿相位与偏振漂移
Heralding	预报	通过辅助探测确认某个光子源确实产生了目标态
TES	Transition Edge Sensor	过渡边缘传感器，早期使用的单光子探测器
PNR	Photon Number Resolving detector	光子数分辨探测器，可区分入射光子个数
SNSPD	Superconducting Nanowire Single-Photon Detector	超导纳米线单光子探测器
QPU	Quantum Processing Unit	量子处理单元
QEC	Quantum Error Correction	量子纠错
Homodyne detection	零差探测	测量光场正交分量的方法，本文中是「做门」的实现手段
FTQC	Fault-Tolerant Quantum Computing	容错量子计算
PennyLane	—	Xanadu 的开源量子编程框架，跨硬件后端